



COMP1850 Programming

School of Computer Science, University of Leeds

Worksheet - Week 1.1

This worksheet contains a combination of formative activities and activities that contribute towards your portfolio.

- Portfolio activities are indicated in the title of the task.
- Activities marked by (*) are advanced activities and may take more time to complete.

Expectations:

1. **Timeliness:** You should complete all the tasks in the order provided and submit your portfolio task to Gradescope before the deadline.
2. **Presentation:** You should present all your work clearly and concisely following any additional guidance provided by the module staff in the module handbook.
3. **Integrity:** You are responsible that the evidence you submit as part of your portfolio evidence is **entirely your own work**. You can find out more about Academic integrity on the Skills@library website. All work you submit for assessment is subject to the academic integrity policy.

Feedback: Feedback on formative activities will be provided via Lab classes and tutorials. Feedback on evidence submitted as part of the portfolio will be available on Gradescope.

Support opportunities: Support with the activity sheet is available in the Lab classes and tutorials. Individual support is available via office hours.

Expected complete date: Friday 9th Oct

Portfolio tasks should be entirely your own work. AI tools (such as Microsoft Copilot) can be used in an assistive role. You are permitted to use AI tools for specific defined processes within the assessment.

Within this assessment you **may** use Generative AI to:

- help you start to understand more complex ideas by providing accessible summaries.
- test your knowledge against a piece of content.
- help identify and correct spelling mistakes and grammatical errors in your work.
- provide feedback and advice on your overall coding style.

You must **not** use Generative AI to:

- produce written content (code, comments or any other content) which you then submit as your work, regardless of whether you make changes to it.
- rewrite or make changes to any of your work.

Learning Outcomes:

- Use industry-standard tools, such as integrated development environments (IDEs) and version control, to collaboratively solve programming problems and manage codebases.
- Develop programs that use the fundamental programming constructs: assignment and expressions, basic I/O, conditional and iterative statements.

Task 1 – Recapping Terminal Commands

You have been introduced to a variety of terminal commands this week – fill out the table with an explanation for each command which says what it is used for.

You may find it useful to run these commands in the terminal of Codespace to remind yourself.

If you would prefer to fill this in online, task1.md contains a markdown version of this table.

Command	Explanation
pwd	
ls	
cd directory_name	
cd ..	
cd -	
mkdir directory_name	
touch filename	
git status	
git add -A	
git commit -m ""	
git push	
git pull	

Task 2 – PORTFOLIO – Basic Python

Pass: You must pass all 3 Gradescope tasks to pass the portfolio.

Minimum standard: Passing the first 2 tests is acceptable but must be corrected to a Pass later on.

You are making an online savings calculator for a bank.

Users will enter the amount of money they want to save every month, and the app will calculate and display:

- The amount they will save every year
- The total amount of money including interest (which will be 0.8% of the final annual savings) they will have saved in a year.

If the user enters a non-integer amount, the app should print out the message 'Invalid amount' and stop running.

A very basic scaffold has been provided in task2.py, and two example outputs are available in example_outputs.txt.

Your code will be automatically graded on Gradescope – you should upload the file with the name 'task2.py' and use the feedback from the grader to make any improvements needed until you pass all the tests.

Advice:

Your file must be named exactly "task2.py".

Start from the template: the correct sequence of print statements must be present.

Task 3 – Exam-style Question

Version control systems such as Git are widely used in software development.

State two advantages of using version control when writing software. Explain each point from:

- the perspective of an individual programmer;
- the perspective of working in a team developing software.

State one disadvantage of Git. You can write this from either:

- your own perspective;
- the perspective of a new programmer who has just been introduced to it.

You may use external resources to help you, but answers must be written in your own words.